

SE250B4 EVB

Hardware User Guide

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1 Applicability Table

Table 1: Applicability Table

Products
SE250B4 SMART EVB MAINBOARD
SE250B4-NA INTERFACE FOR SMART EVB
SE250B4-EU INTERFACE FOR SMART EVB
SE250B4-WE INTERFACE FOR SMART EVB

2 Introduction

2.1 Scope

This document introduces the Evaluation Board (EVB) for the SE250B4 Smart Module, detailing its characteristics and functionalities. The features and solutions outlined in this document apply to all SE250B4 Smart Module variants, as listed in the applicability table.

2.2 Audience

This Telit SE250B4 Smart Module EVB is a development board designed for engineers, programmers and developers for quickly validating the features, peripherals, and develop/debug test applications.

2.3 Contact Information, Support

For technical support and general questions, e-mail:

- TS-EMEA@telit.com
- TS-AMERICAS@telit.com
- TS-APAC@telit.com
- TS-SRD@telit.com
- TS-ONEEDGE@telit.com
- TS-LOCATION@telit.com

Alternatively, use: <https://www.telit.com/contact-us/>

For Product information and technical documents, visit: <https://www.telit.com>

2.4 Conventions

Note: Provide advice and suggestions that may be useful when integrating the module.

Danger: This information MUST be followed, or catastrophic equipment failure or personal injury may occur.

Warning: Alerts the user on important steps about the module integration.

All dates are in ISO 8601 format, that is YYYY-MM-DD.

2.5 Terms and Conditions

Refer to <https://www.telit.com/terms-conditions/>.

2.6 Disclaimer

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3 General Product Description

3.1 Overview

The Telit SE250B4 Smart Module is a multi-mode and multi-band wireless smart module, which is based on Qualcomm QCM2290 platform. It includes baseband, memory, RF front-end and required circuitry to support rich multimedia features, global location-based services, wireless connectivity, and air interface standards including GSM, WCDMA, and LTE.

With higher integration to reduce PCB surface area, time-to-market, and BOM costs, Telit SE250B4 Smart Module will help drive wireless products adoption in industries around the world.

SE250B4 Smart Module is based on Android operating system and it helps customers to develop embedded devices without host MCU or processors.

The SE250B4 Smart Module EVB enables customers to evaluate and test the module's functionalities with a first set of peripherals such a display, camera, and sensors.

The SE250B4 Smart Module EVB includes the following components:

- LCM
- HDMI, LVDS
- Proximity Sensor and Ambient Light sensor
- Accelerometer (G-Sensor) and Gyroscope
- Three Cameras (2MP, 5MP, 13MP)
- Analog and digital audio (RJ9 handset connector, 3.5mm audio jack, onboard analog speaker, and digital microphones for noise reduction)
- Dual SIM card holders
- One USB (3.1) interface for the modem to issue AT commands and firmware upgrading. This can also be used to charge the battery.
- One USB (2.0) interface connected to USB-UART converter to communicate with two UARTs (debug UART and UART1) of the module.
- 4 push buttons (Power Key, Reset, VOL+ and VOL-)
- TF card slot
- DC jack with Battery charging circuit
- Vibrator

3.2 Kit Contents

The SE250B4 Smart Module EVB kit includes the following components:

Table 2: Kit Contents

Item	Quantity
Telit SE250B4 Smart Module EVB Mainboard	1
5V DC Power Supply adapter	1
TLB1(SE250B4 module and RF connectors Board)	1
TLB2(Touchscreen and Camera Board)	1
Multi Band RF antennas with SMA male connector	4

The image below shows the EVB Mainboard assembled with TLB1 and TLB2 boards.

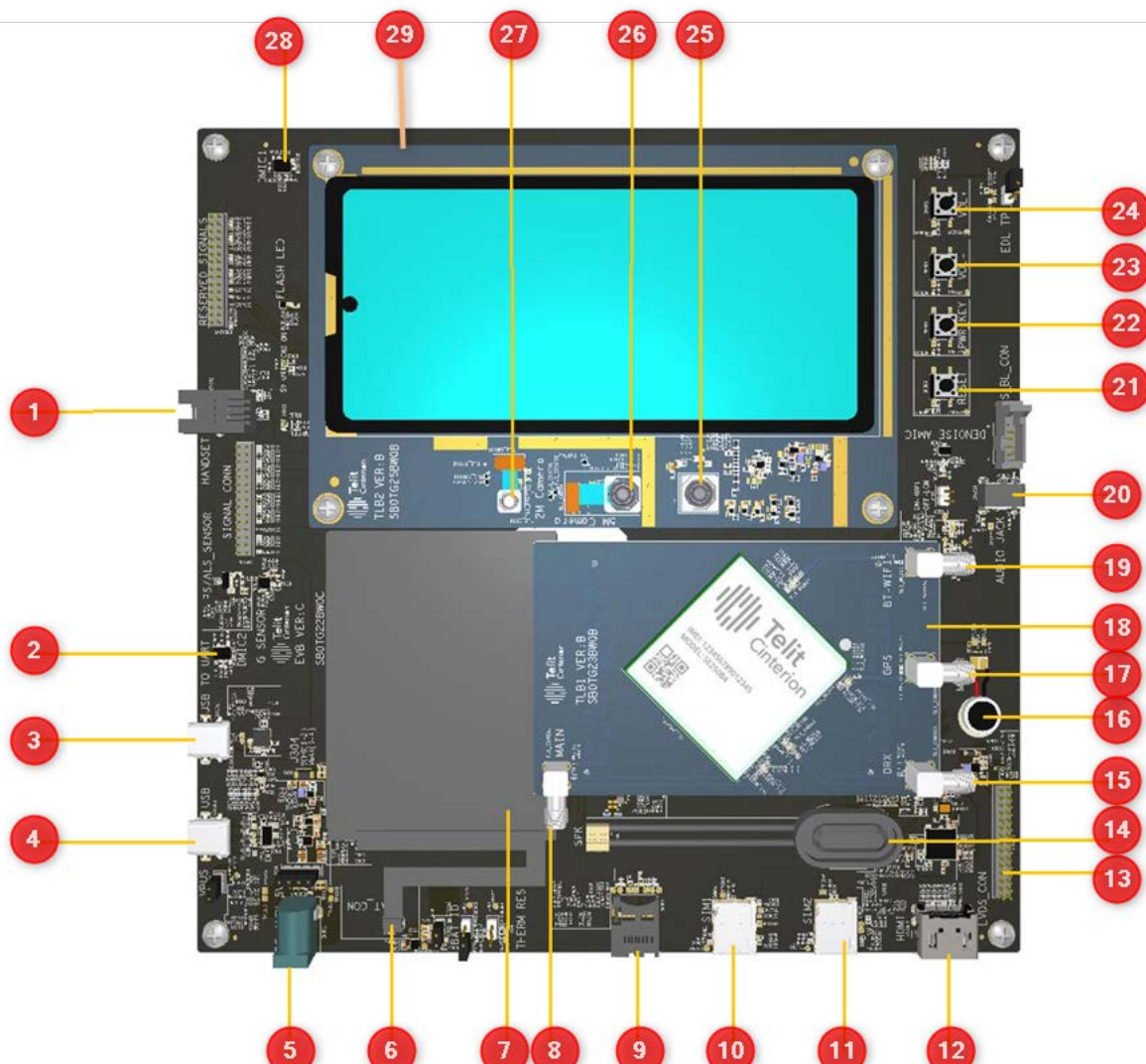


Figure 1: EVB Mainboard

Table 3: Kit Contents Details

No.	Indicates to	No.	Indicates to	No.	Indicates to
1	RJ9 Handset Connector	14	Speaker	27	2MP Camera
2	DMIC2	15	Diversity Ant SMA Connector	28	DMIC1
3	USB-UART	16	Vibrator	29	TLB2
4	USB	17	GPS Ant SMA Connector		
5	DC Jack	18	TLB1		
6	Battery Connector	19	WiFi-BT Ant SMA Connector		
7	Battery	20	Audio Jack		
8	Main Ant SMA Connector	21	RESET Button		
9	SD Card Connector	22	PWRKEY Button		
10	SIM1	23	VOL- Button		
11	SIM2	24	VOL+ Button		
12	HDMI Connector	25	13MP Camera		
13	LVDS Connector	26	5MP Camera		

The **TLB1** board as shown below, includes the SE250B4 Smart Module with RF antenna connectors:

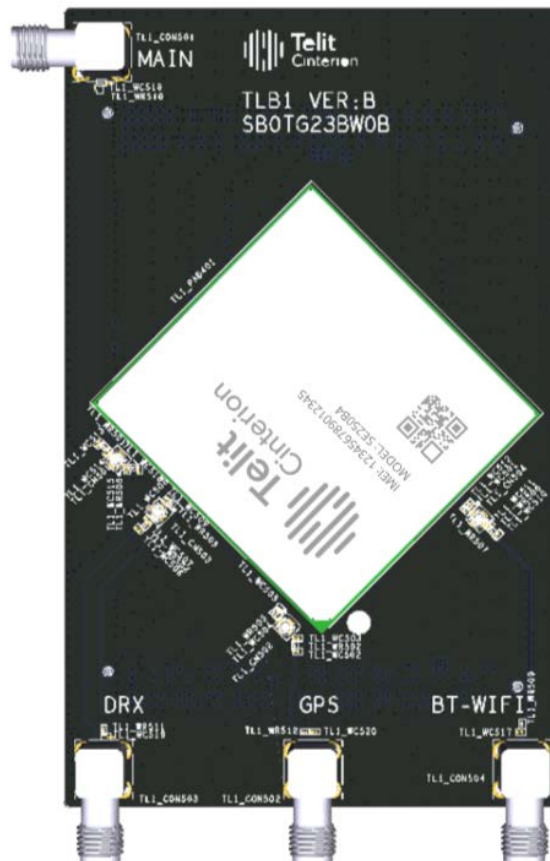


Figure 2: TLB1 Board

The **TLB2** board as shown below, includes the LCM module and three cameras (2MP, 5MP, 13MP):

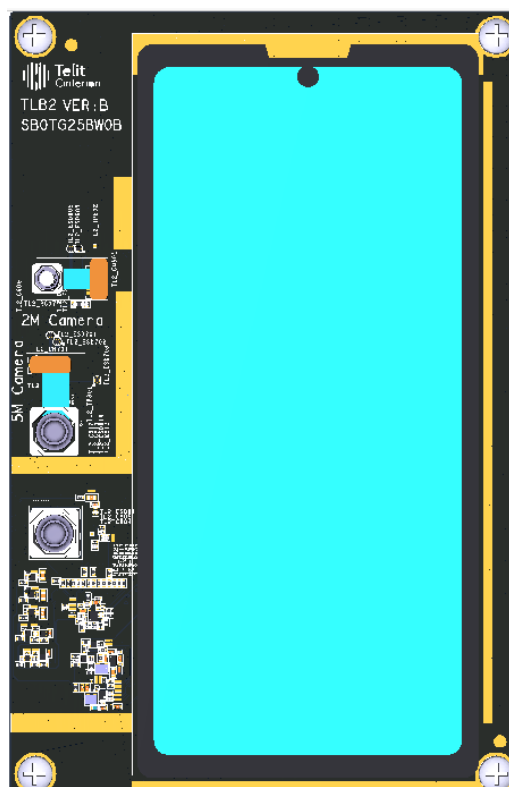


Figure 3: TLB2 Board

4 Functional Description

4.1 EVB Block Diagram

The EVB block diagram below illustrates the various peripherals, sensors, connectors, and their connections to the SE250B4 Smart Module.

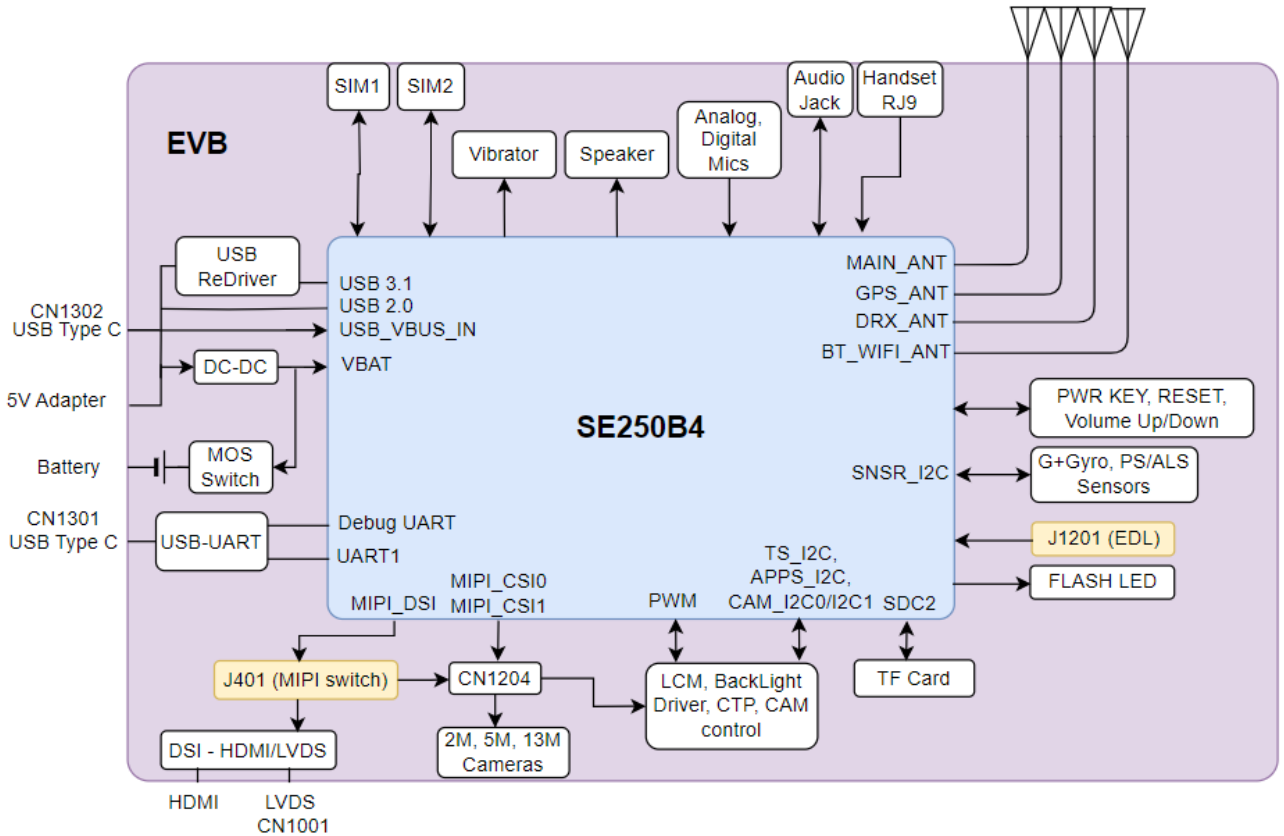


Figure 4: SE250B4 EVB Block Diagram

4.2 Default Jumper Settings

The image below displays the main jumpers locations on the EVB.
For default jumper settings, please refer to [Table 3](#).



Figure 5: Jumpers locations on EVB

The table below describes the purpose and use of each jumper on the EVB:

Table 4: Default Jumper Settings Description

Jumper / Switch	Default Setting	Function	Statuses
J305	Open	Battery Simulation	Place a jumper to simulate the battery ID
J301	Closed	Simulate Battery NTC	Place a jumper to simulate the battery NTC
J302	Closed	VBATT	Place a jumper to supply VBATT to the Smart Module
J304	Closed [pins 3-4]	DC Power Source	Place jumper on pin 3 and 4 for selecting power source from DC adaptor. To change power source to USB, place two jumpers (1-2 and 3-4). Note that power adapter shall be removed from Jack in this case and J303 must be closed.
J303	Open	USB Power source	Place a jumper to switch the power source to USB (remove DC adaptor)
J401	Open	HDMI-LCM Selection	Open: LCM Closed: HDMI
J1201	Open	USB Force Boot	Place a jumper to enter emergency download mode

5 Power Supply

5.1 Power Supply Source Selection

The EVB can be powered either by a power adapter or a battery, depending on the use case. The following diagram illustrates the power supply source selection:

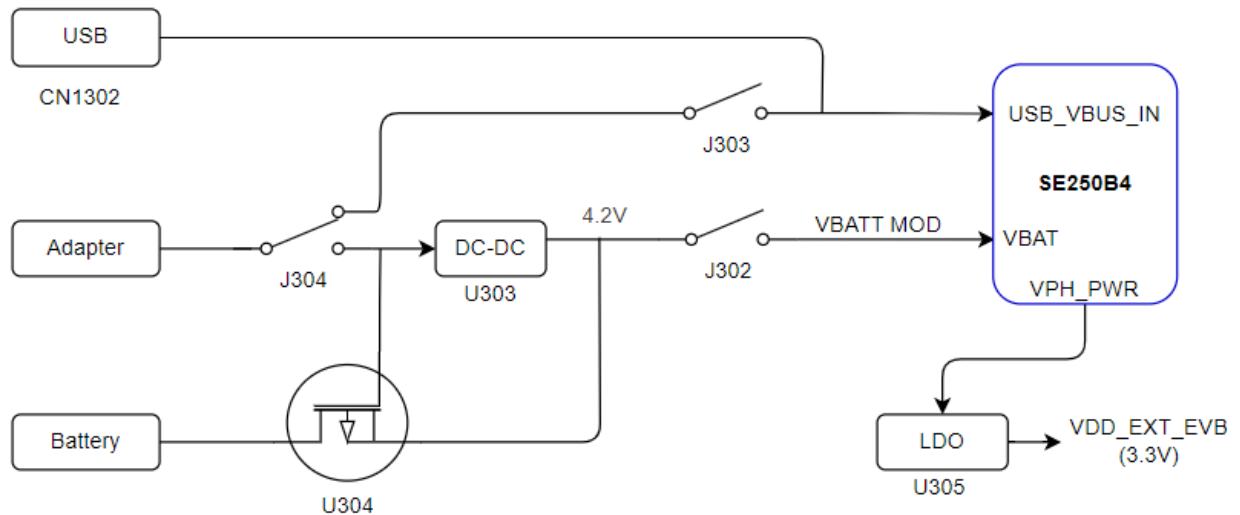


Figure 6: Power Supply Source Selection

- 1 **Power up EVB from 5V power adapter:** Jumper must be set to 3-4 pins on J304, and jumper on J302.
- 2 **Power up EVB from USB (CN1302):** J303 and J302 must be closed. J304 must have jumpers on pins 1-2 and 3-4. Ensure the power adapter is disconnected in this case.

The battery remains connected to both power sources. MOSFET U304 prevents the power adapter and battery from supplying power simultaneously.

There is a Power-on LED indicator (D302) on the EVB, which glows green when the power supply from the DC adapter is present. RED LED blinks if the USB is connected and battery is too low to boot. RED LED stops blinking after boot and normal operation.

6 Peripherals

6.1 Display

The display signal routing on the EVB is illustrated in the block diagram below. MIPI DSI interface signals are brought to MIPI switch IC. This switch routes the DSI signals to LCM or HDMI/LVDS IC based on the jumper J401 setting.

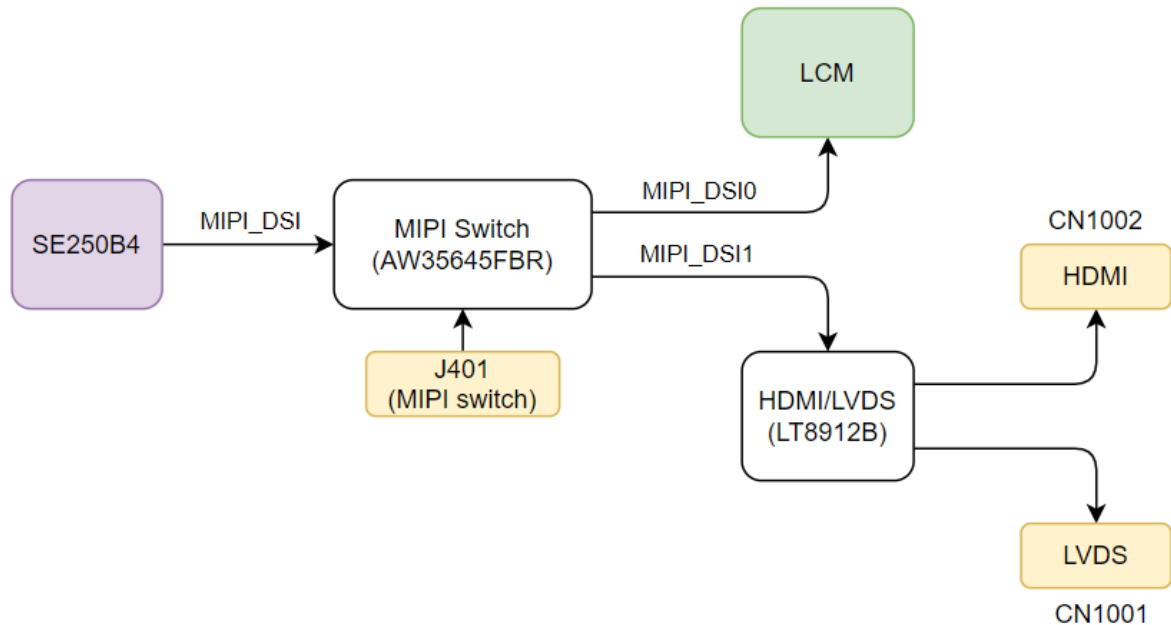


Figure 7: EVB Display Signal Routing

6.1.1 LCM

The primary peripheral of the Smart Module is the Display (also called LCM) , which includes additional items such as the Capacitive Touch Panel (CTP) and a backlight driver. The specifications for the LCM used with this EVB are as follows:

Table 5: LCM specification

Item	Description
Display Type	TFT-LCD/Transmissvie
Resolution	720*1520
Pixel Pitch	0.02875X3(H)*0.08625(V)
LCM Driver IC	FL7703N-G5-DP
Display Mode	Transmissive /Normally Black
LCM Interface Type	MIPI Interface(D-PHY), TP:I2C
Pin Pitch	0.4Pitch
Input Voltage	1.8V and +-5.5V; TP:3V
Backlight Type and Colour	LED backlight /white
Backlight	15 LED, 3 groups in parallel, 5 LEDs in series in each group, Vf=15.25V, If=3*22mA

The LCM is located on the TLB2 board, which is connected to the EVB through the B2B connector CN1204.

6.1.2 HDMI

HDMI signals from the LT8912B are routed to the HDMI connector CN1002. Jumper J401 must be placed to route the display signals to HDMI.

6.1.3 LVDS

LVDS signals from the module are routed to the 30-pin connector CN1001. The MIPI_DSI interface routing can be seen in the block diagram.

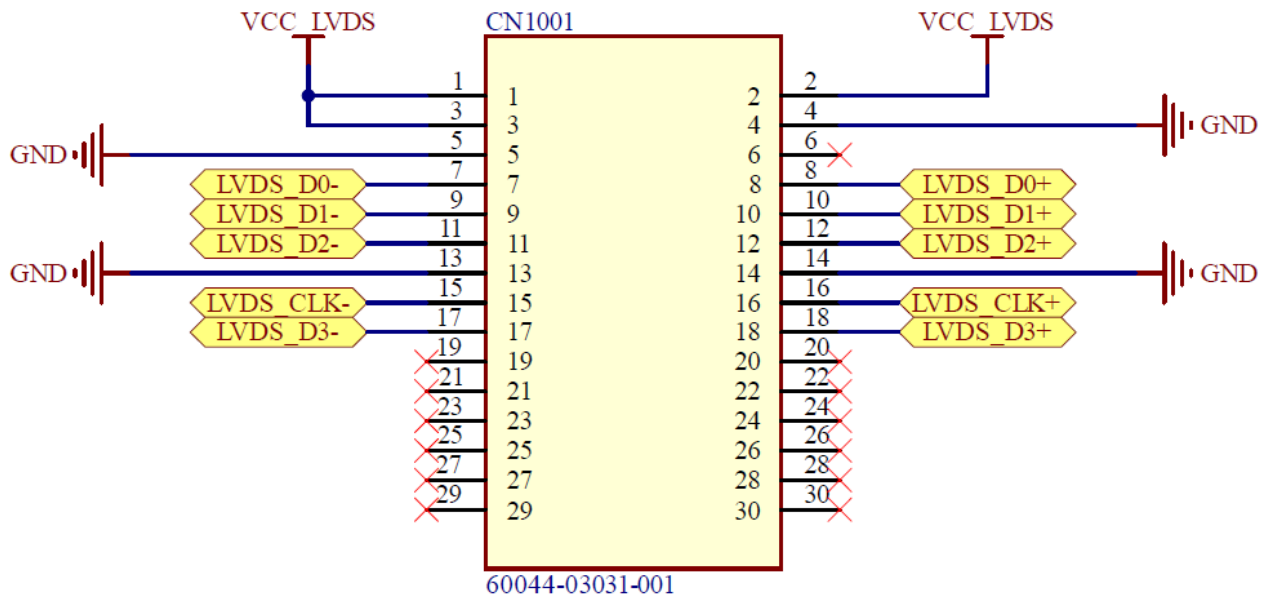


Figure 8: LVDS Signals

6.2 USB

To download firmware to the module and debug/trace there is a USB type C connector (CN1302) on EVB. Battery can only be charged from this port.

6.3 USB-UART

The module has two UART ports; Debug UART and UART1 (with flow control pins CTS and RTS). The second UART port is for the debug. Both these UARTs are connected to UART to USB converter CP2105. When the user connects a Type-C USB cable to CN1301, two COM ports will be visible in Windows Device Manager.

6.4 SIM Interfaces

The EVB has two SIM slots. Both slots support SIM presence pin and support Hot plug.

- SIM1 is on connector CN902.
- SIM2 is on connector CN903.

6.5 SD CARD

EVB has a micro SD Card slot at CN901 and supports Hot plug.

6.6 Control Buttons

EVB has four push buttons. These buttons functions are explained in the sub sections below.

6.6.1 Power

The Power Key is used to power on and off the SE250B4 Smart Module.

To power on the Module:

- 1 Connect the power adapter
- 2 Switch on the power supply.
- 3 Press and hold the POWER Key till the Telit Logo is displayed on LCM.

For more information, refer to the SE250B4 Smart Module Hardware Design guide.

The following is the POWER key circuit:

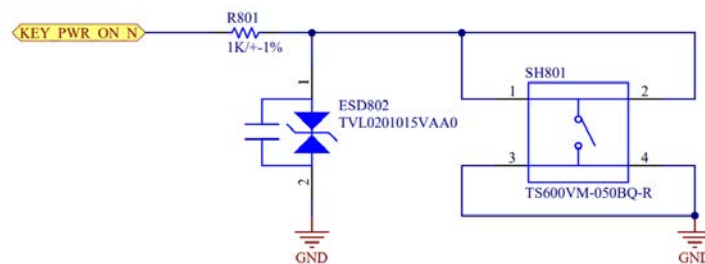


Figure 9: Power Button Circuit

6.6.2 Reset

The RESET button is used for unconditional reboot of the SE250B4 Smart Module. For the detailed timings, refer to the SE250B4 Smart Module Hardware Design guide.

The following is the RESET button circuit diagram:

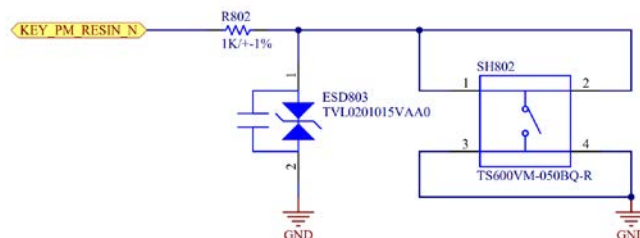


Figure 10: Reset Circuit

6.6.3 Volume

There are two buttons for the audio volume control.
The following is the volume up circuit diagram:

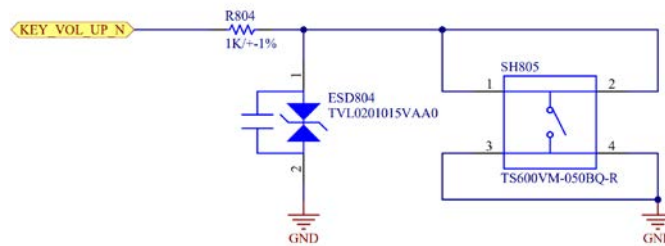


Figure 11: Volume Up Key Circuit

The following is the volume down circuit diagram:

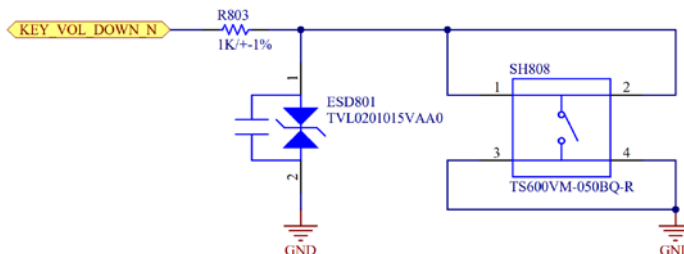


Figure 12: Volume Down Key Circuit

6.7 Sensors

The EVB has two sensors:

- **ICM-42670 from TDK:** It is a 6-axis sensor with a 3-axis gyroscope and a 3-axis accelerometer in a single package.
- **MN78911DKDN from Eminent:** It is an ALS and Proximity sensor that contains both ambient light sensing and proximity detection in a single package.

6.7.1 ICM-42670 (Accelerometer and Gyroscope)

This 6-axis sensor integrates a 3-axis gyroscope and a 3-axis accelerometer into a single package.

Table 6: Sensor Details

Item	Value
Serial interface	I2C
I2C 7-bit Address	0x68

6.7.2 MN78911DKDN (ALS and Proximity)

This ALS and proximity sensor integrates ambient light sensing and proximity detection in a single package.

Table 7: Proximity Sensor

Item	Value
Serial interface	I2C

Item	Value
I2C 7-bit Address	0x47

6.8 Audio

The EVB is equipped with a 3.5mm audio jack, RJ9 handset, onboard speaker, digital microphones (DMIC1, DMIC2), and De-noise analog Mic.

6.8.1 Handset

The RJ9 Handset connector is located at CN602. The handset is often used for audio testing.

The RJ9 pinout is shown in the table below:

Table 8: Handset Pinout Description

Pin	Signal
1	MIC_M
7	MIC_P
3	EAR_P
5	EAR_M

In the hands-free mode, the MIC used is the handset MIC, and the speaker is switched to the one present on EVB.

6.8.2 Headset

The EVB has 3.5mm audio jack (CON601), supports earphones having detection pins HS_DET and HPH_L (detection circuit).

The following is the Headset jack pinout diagram:

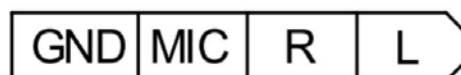


Figure 13: Headset Description

When the headset is inserted, an interrupt triggers the SE250B4 Smart Module to switch the audio lines to headset mode.

6.8.3 Speaker

The EVB features an onboard speaker at CN601 for hands-free mode. The audio to speaker is amplified by the class K audio amplifier AW8737AFRCR.

6.8.4 Vibrator

The EVB includes a vibrator connected to TP601 and TP602.

6.9 ADC

The two ADC's are connected to a voltage divider on the EVB as shown in the circuit below. Note that R710 and R711 are not mounted. These two resistors (resistance value as required) must be mounted to test the ADCs to read the voltage, otherwise, the ADC reading will be Zero volts. For more details on the ADC range and spec, refer to the SE250B4 Smart Module Hardware Design guide.

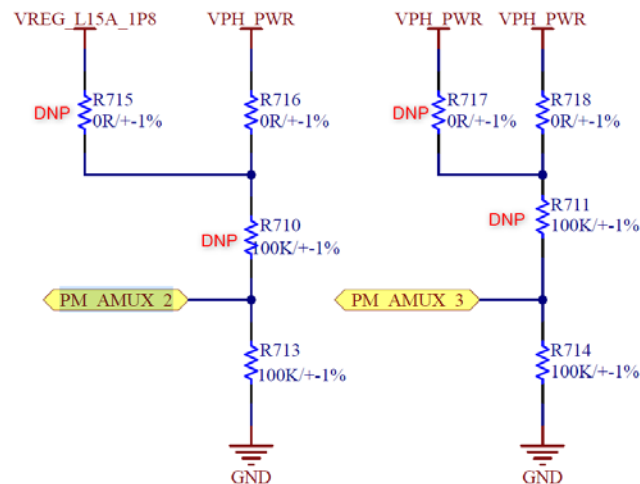


Figure 14: ADC Circuit

6.10 VCOIN

VCOIN serves as the backup battery for RTC. This is located on the EVB, below the TLB1 board. It can be soldered to the SMD pads of BAT701.

The following is the circuit diagram:

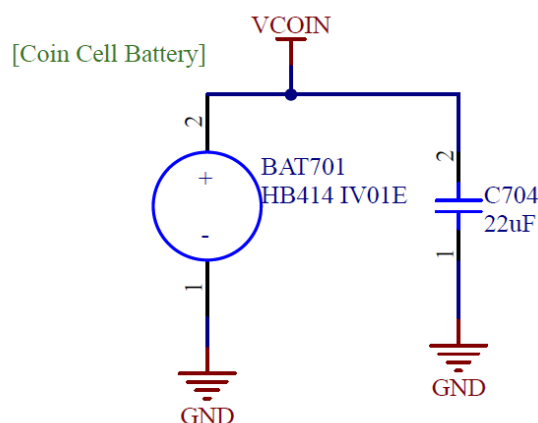


Figure 15: VCOIN Circuit

6.11 Signal Connectors

There are two 30-pin connectors named CN1201 and CN1205. These are 2mm pitch through-hole connectors. Peripherals UART1, UART2, Debug UART, LCM, ADC, SPI, I2C, and GPIOs can be accessed at these connectors. Note that there are also a few other signals brought to these connectors.

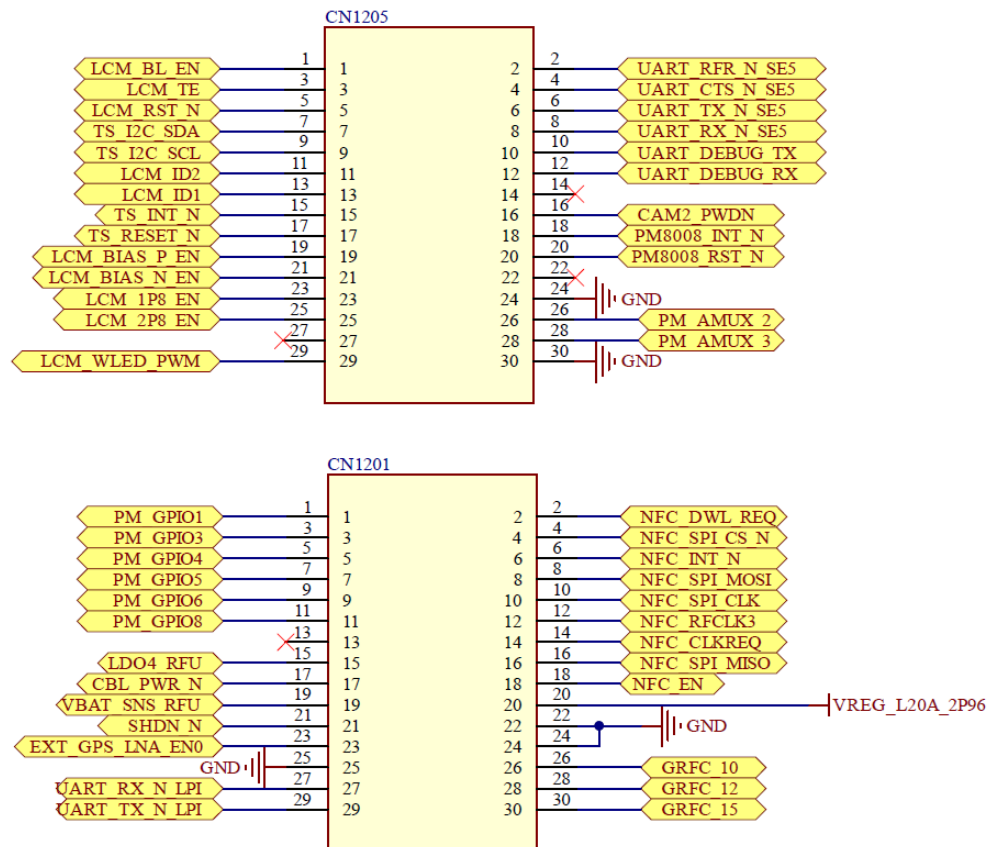


Figure 16: Signal Connectors

6.12 USB Boot

To perform a USB boot, place the jumper J1201 on the EVB. This will make the FORCE USB BOOT signal high. By default, this jumper is not mounted.

The following is the circuit diagram:

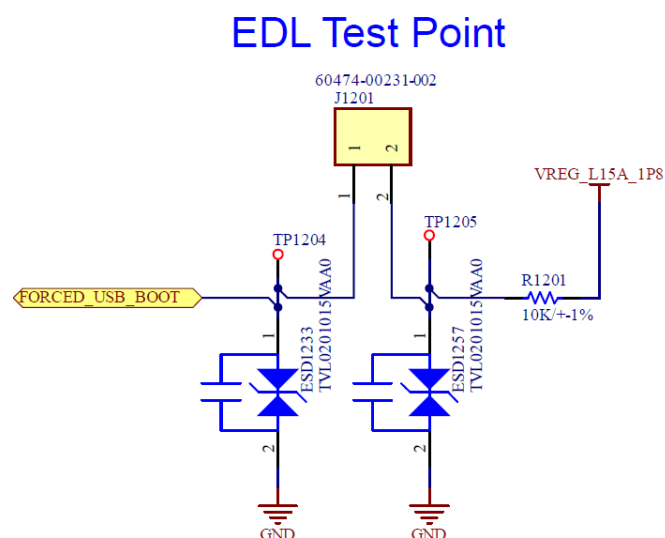


Figure 17: USB Boot Circuit

7 B2B Connectors

The Module's Interface (TLB1) and LCM Display TLB (TLB2) can be connected to the EVB through three socket connectors from SAMTEC with p/n SEAF-40-05.0-L-04-2-A-K-FR (1600 PIN 0.5mm pitch) .

7.1 TLB1

The TLB1 is connected to the EVB through two header connectors located on the bottom side, both from SAMTEC, part number SEAF-40-05.0-L-04-2-A-K-FR (1600 PIN, 0.5mm pitch). These connectors are labelled TL1_CN301 and TL1_CN302.

This interface board includes four RF connectors: main antenna, Diversity antenna, GPS antenna and Wi-Fi/BT antenna. To use antennas other than those provided in the kit, refer to the SE250B4 Smart Module Hardware Design guide for the antenna specifications.

Note: It is recommended to use passive antennas. Active antennas can increase the signal, but an excessive gain can blind the receiver. Make sure to follow the RF routing guideline for RF track routing.

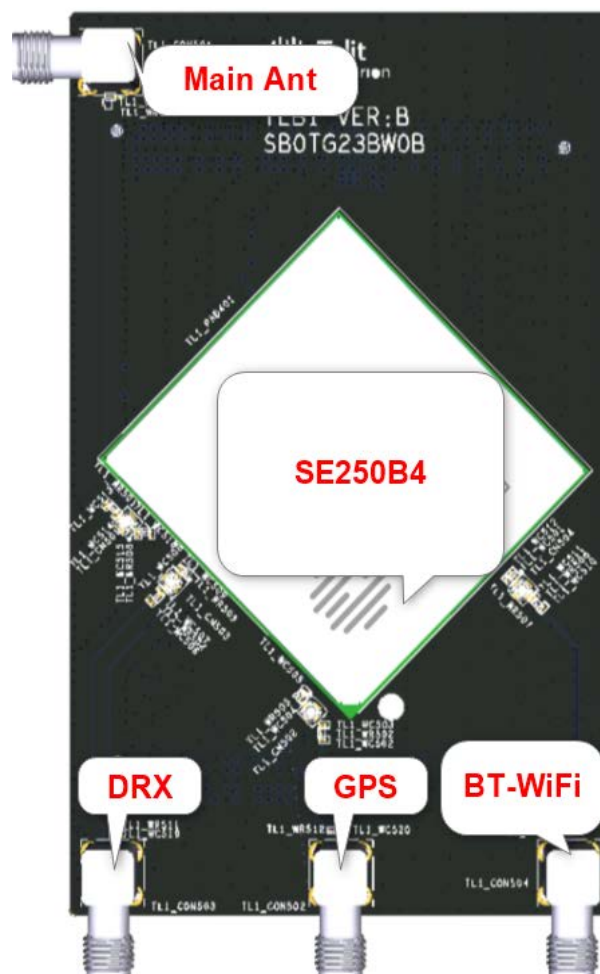


Figure 18: TLB1

7.2 TLB2

The TLB2 board, which includes the LCM (Display) and cameras, connects to the EVB via the B2B connector TL2_CN301 located on the bottom side of the TLB2. This connector is a SAMTEC part with p/n SEAF-40-05.0-L-04-2-A-K-FR (1600 PIN, 0.5mm pitch), assembled on the bottom of the PCB.

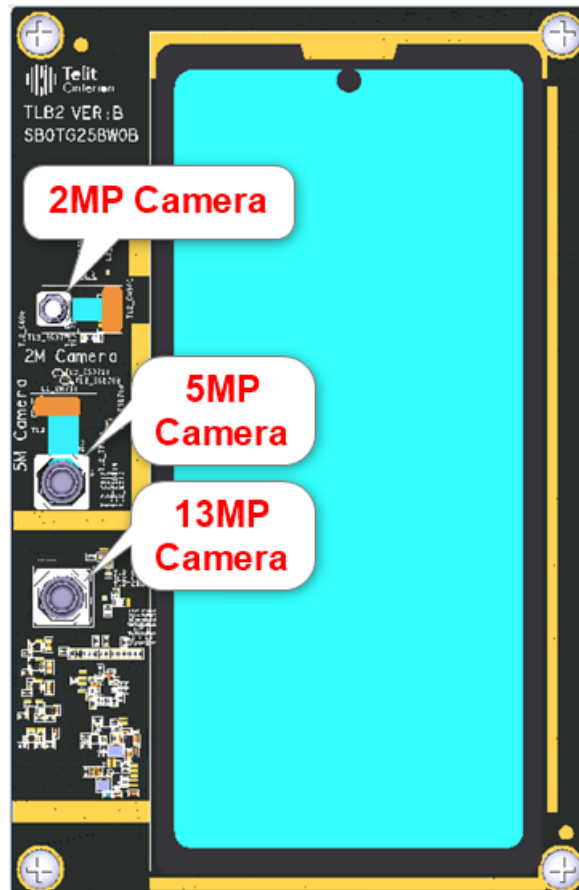


Figure 19: TLB2

The TLB2 board features three camera connectors. For guidance on assembling the camera modules onto the TLB2, refer to the [Figure 19](#).

7.2.1 Front Camera

The front camera is a 2MP sensor using the KPFC228 IC, with an I2C address of 0x7C(W)/0x7D(R).

7.2.2 Rear Camera

The rear cameras consist of 5MP and 13MP sensors, utilising the CMF008-B5BA-E and AAD12B ICs, respectively.

Table 9: Rear Camera I2C Addresses

	Sensor I2C address	Driver I2C address
CMF008-B5BA-E	0x40 (W)/ 0x41 (R)	0x18 (W) / 0x19 (R)

8 Telit Evaluation Board (EVB) Schematics Discrepancies

The EVB schematics shared with the customer contain a few noticeable discrepancies, as listed below. Kindly review these observations and incorporate the recommended changes into your schematic design, where applicable.

S/N	Component	Observation and comments
1	EVB	Some of the components mentioned in schematics are not populated and are not marked as DNP (Do Not Populate), which may lead to some confusion. Please reach out to Telit support(TS-SRD@telit.com) for queries and information.
2	CN1202B	Pins D7 and D8 are expected to be connected to USB_VBUS_IN. However, D8 and D9 are connected to USB_VBUS_IN, while D7 is connected to GND.
3	CN1202B	Pin D30 should be connected to UART_CTS_N_SE5, whereas the signal is currently routed to D31.
4	CN1202B	Pin D31 should be connected to UART_RX_N_LPI. Instead, the signal is connected to D32, while D32 should be connected to GND.
5	CN1202B	Pin C23 is expected to remain floating, but is currently connected to GND.
6	CN1202B	Pin C12 should be connected to GND, but is currently left floating.
7	CN1204A	Signal VREG_S3A_1P352 is connected to A26, whereas it should be connected to A27.
8	CON601	VREG_L15A_1P8 should be connected to CDC_HS_DET. Instead, it is connected to the left headphone output (CDC_HPH_L).
9	R341	Resistor value is shown as 10 kΩ, whereas the expected value is 100 kΩ.
10	R342	Resistor value is shown as 10 kΩ, whereas the expected value is 100 kΩ.

9 Acronyms and Abbreviations

Table 10: Acronyms and Abbreviations

Acronym	Definition
ALS	Ambient Light Sensor
B2B	Board-to-Board Connector
CTS	Clear to Send
DMIC	Digital Microphone
DSI	Display Serial Interface
EN-DC	E-UTRA – NR Dual Connectivity
GPIO	General Purpose Input Output
HDMI	High-Definition Multimedia Interface
HS	High Speed
I/O	Input Output
I2C	Inter-integrated Circuit
LCM	Liquid Crystal Module (Display)
LTE	Long Term Evolution
LVDS	Low Voltage Differential Signaling
MIPI	Mobile Industry Processor Interface
NTC	Negative Temperature Coefficient
PCB	Printed Circuit Board
RTS	Request to Send
SIM	Subscriber Identification Module
SOC	System-on-Chip
SMPS	Switching Mode Power Supply
UART	Universal Asynchronous Receiver Transmitter
USB	Universal Serial Bus
WDM	Wideband Code Division Multiple Access

10 Related Documents

Refer to <https://dz.telit.com/> for current documentation and downloads.

Table 11: Related Documents

S.no	Doc Code	Document Title
1	1VW0301853	SE250B4 Evaluation Board Quick Start Guide
2	1VW0301817	SE250B4 Smart Module Hardware Design Guide
3	1VW0301823	SE250B4 SDK User Guide

11 Document History

Table 12: Document History

Revision	Date	Changes
1	2026-07-23	Highlighted J1201 jumper in Figure 5 Added section 8 mentioning all discrepancies in EVB schematics
0	2024-10-16	Initial Release

From Mod.0818 Rev.11

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